IAT New Test 2

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

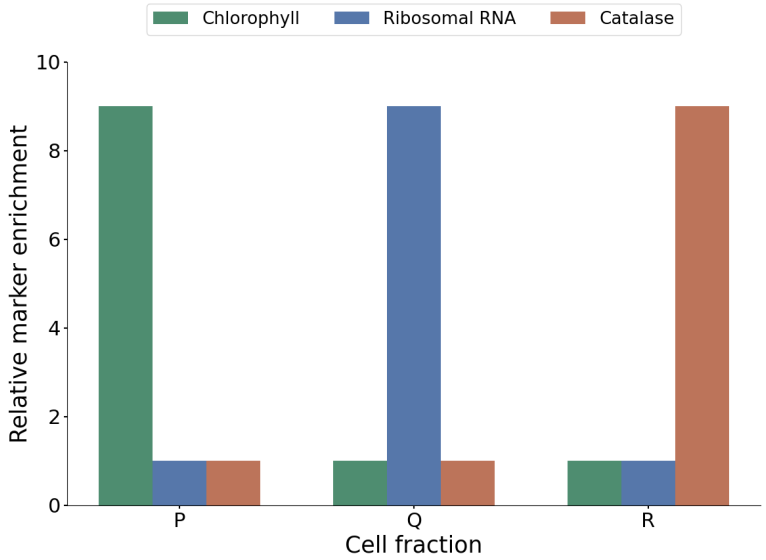
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A seed plant bears ovules exposed on cone scales and does not enclose its mature seeds in a fruit. Which group fits both observations?
A) Gymnosperms
B) Angiosperms
C) Bryophytes
D) Pteridophytes

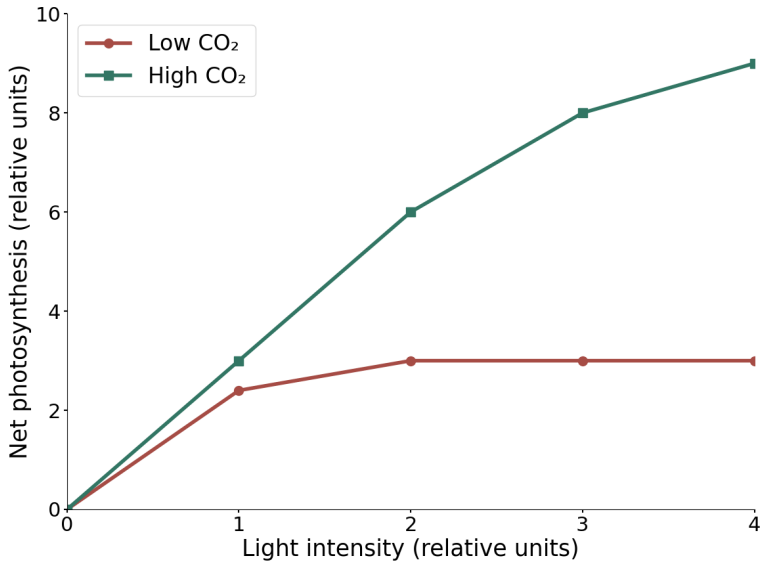
Q2 In a flower, sepals, petals and stamens appear above an ovary to which the floral tube is fused. Which pair describes it?
A) Superior ovary; hypogynous
B) Inferior ovary; epigynous
C) Superior ovary; perigynous
D) Inferior ovary; hypogynous

Q3 The figure compares three cell fractions after independent repeat isolation. Which fraction most strongly supports a peroxisome identification, using both positive and negative markers?



A) Fraction P, because chlorophyll is abundant
B) Fraction Q, because ribosomal RNA is abundant
C) Fraction R, because catalase is enriched while chlorophyll and ribosomal RNA are low
D) All three equally, because each contains protein

Q4 The figure plots net photosynthesis against light at two fixed CO_2 concentrations and constant temperature. Which conclusion is best supported over the measured range?



A) Extra light always lowers net photosynthesis
B) High CO_2 prevents any light response
C) The two curves show that temperature changed
D) The low-CO_2 curve plateaus sooner, while high CO_2 permits a larger response to light

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Q5 At rest, heart rate is 60 min^{-1} and stroke volume is 80 mL. During exercise, heart rate becomes 100 min^{-1} while cardiac output rises by 25%. What is the new stroke volume?
A) 60 mL
B) 80 mL
C) 100 mL
D) 120 mL

Q6 A species has 2n=12. One homologous pair fails to separate in meiosis I of a male meiocyte; meiosis II is normal. Each of the four sperm is considered for fertilisation with a normal egg. Which zygote chromosome counts can result?
A) Only 12
B) 11 and 13
C) 10 and 14
D) 12 and 13

Q7 Crossing two pure lines produces F_1 petals with distinct red and white patches; assays detect both parental pigment types. Which inheritance pattern best fits the observations?
A) Complete dominance of red
B) Incomplete dominance producing a new pink pigment
C) Codominance of the two pigment traits
D) Sex-linked recessive inheritance

Q8 Equal doses of one antigen are given 40 days apart. Antibody rises slowly after the first dose but rapidly and much more after the second. Which mechanism explains the change without assuming a higher second dose?
A) The second injection bypasses antigen recognition
B) All innate immune cells become antigen-specific antibodies
C) The antigen becomes a different molecule after 40 days
D) Memory lymphocytes formed after first exposure respond faster on re-exposure

Q9 A vector and insert are cut with the same restriction enzyme to produce complementary overhangs. A second vector has blunt ends. All ends have the groups required by ligase. What advantage is specific to the first vector–insert pair?
A) Complementary overhangs can anneal and align ends before covalent ligation
B) Ligase acts only on overhangs and never on blunt ends
C) Complementary ends ensure a single insert orientation in every case
D) Cutting makes both DNA fragments single-stranded throughout

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Q10 At equal intervals, a closed population is counted as 50, 90, 140, 180, 200 and 205; food supply is fixed and counts are accurate. Which interpretation fits the changing increments?
A) Exponential growth at a fixed per-capita rate
B) Growth slows as limiting resources become important
C) The population must have declined between every count
D) Immigration alone explains the final plateau

Q11 An unaffected person has an affected sibling with an autosomal recessive disorder; both parents are unaffected. The person's unaffected partner has one confirmed carrier parent and one confirmed non-carrier parent. Under complete penetrance and independent families, what is their first child's affected risk?
A) 1/24
B) 1/8
C) 1/12
D) 1/6

Q12 An mRNA coding region reads AUG | GAA | UCC | GCU | UAA. A single U is inserted immediately before the first base of the second codon, with translation still starting at AUG. What follows without consulting a codon chart?
A) Only one amino acid can change
B) Codon boundaries after AUG shift and several downstream amino acids may change
C) The start codon must disappear
D) Every codon after insertion remains unchanged

Q13 A freely filtered solute has plasma concentration 1 mg/mL and filtration rate 100 mL/min. Urine flows at 1 mL/min with solute concentration 10 mg/mL. With no tubular secretion, what fraction of the filtered solute is reabsorbed?
A) 1%
B) 10%
C) 50%
D) 90%

Q14 A drug prevents spindle microtubules from attaching to kinetochores but leaves DNA replication intact. Cells accumulate with condensed chromosomes and seldom separate chromatids. Which immediate process is blocked?
A) DNA replication in S phase
B) Nuclear envelope breakdown before spindle formation
C) Anaphase separation of sister chromatids
D) Cytoplasmic translation of spindle proteins

Q15 Two C_3 leaves are held at identical light, temperature and stomatal conductance. Leaf Q has higher O_2 and lower CO_2 around Rubisco than leaf P. What change is expected in Q?
A) More oxygenation and photorespiratory carbon loss
B) No change because Rubisco binds only CO_2
C) More net fixation because O_2 is the carbon source
D) Elimination of the Calvin cycle

Chemistry

Q16 Why is the H–N–H angle in NH_3 smaller than the H–C–H angle in CH_4?
A) Nitrogen forms four identical N–H bonds
B) Methane has a lone pair on carbon
C) Ammonia contains no tetrahedral electron-pair arrangement
D) A nitrogen lone pair repels bonding pairs more strongly than a bonding pair does

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Q17 The figure shows complexes P and Q with differently placed bromide and sulfate. Assume only freely dissociated counterions give immediate precipitates. Which observation distinguishes them on separate $BaCl_2$ and $AgNO_3$ tests?

P

$$[Co(NH_3)_5Br]\ SO_4$$

Q

$$[Co(NH_3)_5SO_4]\ Br$$

A) P gives $BaSO_4$ and Q gives AgBr
B) P gives AgBr and Q gives $BaSO_4$
C) Both give each precipitate immediately
D) Neither gives an immediate precipitate

Q18 Hydrogen has $E_n = -13.6/n^2$ eV. What minimum photon energy ionises an atom initially at $n=2$?
A) 13.6 eV
B) 3.4 eV
C) 10.2 eV
D) 1.51 eV

Q19 A primary carbocation lies next to a carbon bearing a hydride and two alkyl groups. If a 1,2-hydride shift precedes nucleophile capture, what chiefly favours that shift?
A) It necessarily creates an aromatic ring
B) It converts the positive charge into a negative charge
C) It relocates positive charge to a more substituted, more stable carbocation centre
D) It breaks all carbon–carbon bonds

Q20 Propene reacts with HBr first without and then with peroxide, with the radical pathway effective in the second experiment. What major products are expected respectively?
A) 1-bromopropane; 2-bromopropane
B) 1-bromopropane; 1-bromopropane
C) 2-bromopropane; 2-bromopropane
D) 2-bromopropane; 1-bromopropane

Q21 An optically active 2-bromobutane sample is divided. P undergoes clean backside SN2 substitution; Q is heated with a small alcoholic base and gives predominantly E2. Which combined outcome fits both mechanisms?
A) P inverts at C-2; Q mainly gives trans-but-2-ene among alkene isomers
B) P retains configuration; Q gives only but-1-ene
C) P racemises completely; Q gives only but-1-ene
D) P inverts; Q gives mainly butan-2-ol

Q22 Liquids X and Y each release gas with sodium metal. Only X gives the usual coloured complex with neutral $FeCl_3$. Which assignment is best supported?
A) X ethanol; Y phenol
B) X phenol; Y a simple aliphatic alcohol
C) X ether; Y alkane
D) X alkane; Y phenol

Q23 Ethanal is treated with dilute base at low temperature, then the isolated aldol product is warmed under dehydrating conditions. What is the pair of major carbon skeletons before and after dehydration?
A) 2-hydroxybutanal; butanal
B) 3-hydroxypropanal; prop-2-enal
C) 3-hydroxybutanal; but-2-enal
D) 3-hydroxybutan-2-one; but-1-ene

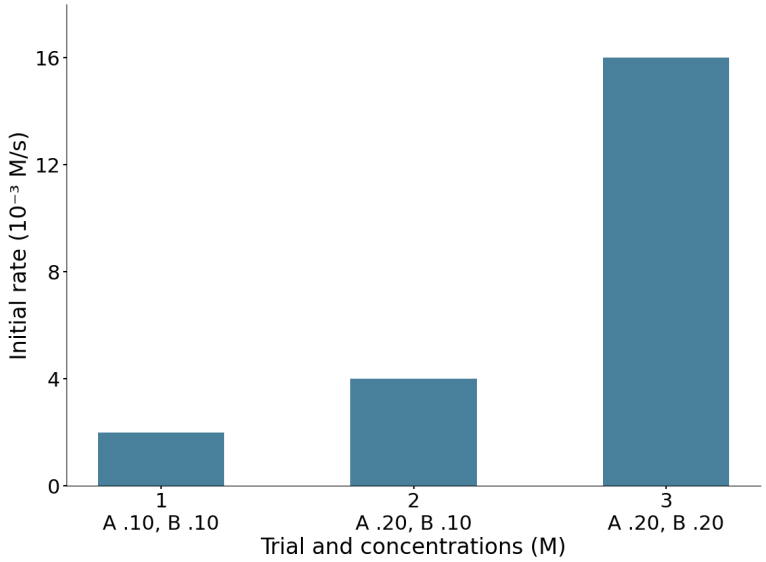
Q24 In water, aniline is appreciably less basic than cyclohexylamine. Which explanation addresses electron-pair availability at nitrogen?
A) Aniline has no nitrogen lone pair
B) Cyclohexylamine is an amide
C) The aniline nitrogen has a different valence
D) Aniline's lone pair delocalises into the benzene ring and is less available for protonation

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Q25 At fixed temperature, 0.020 mol glucose and 0.030 mol urea make 1.00 L ideal solution. Both are nonelectrolytes. What is its osmotic pressure divided by that of 0.020 M glucose at the same temperature?
A) 2.5
B) 1.5
C) 0.4
D) 5.0

Q26 Ag/Ag^+ half-cells at 298 K contain 0.010 M and 1.0 M Ag^+. With ideal solutions and negligible junction potential, use 0.059 V for 2.303RT/F. Which electrode is negative, and what emf results before appreciable composition change?
A) Concentrated; +0.118 V
B) Dilute; +0.118 V
C) Dilute; +0.059 V
D) Concentrated; −0.118 V

Q27 The figure plots initial rates for trials 1–3 while one concentration is doubled at a time. Which empirical rate law follows?



A) k[A]²[B]
B) k[A][B]
C) k[A][B]²
D) k[A]²[B]²

Q28 On the same mole basis, ΔH(A→B)=+40 kJ/mol and ΔH(B→C)=−75 kJ/mol. Find ΔH(C→A).
A) −115 kJ/mol
B) −35 kJ/mol
C) +115 kJ/mol
D) +35 kJ/mol

Q29 20.0 mL of 0.10 M HCl is titrated with 0.10 M NaOH. At 30.0 mL added, find pH at 25 °C, with additive volumes and Kw=10^{-14}.
A) 12.30
B) 1.70
C) 7.00
D) 13.30

Q30 A transition-metal ion has exactly one unpaired electron. Under the spin-only approximation, what magnetic moment in Bohr magnetons is predicted?
A) 1
B) √3
C) √8
D) 3

Mathematics

Q31 A circle with centre (3,4) touches the x-axis. Which ordinates occur where it intersects the positive y-axis?
A) $4+\sqrt{7}$ only
B) $4-\sqrt{7}$ only
C) $4\pm\sqrt{7}$
D) No intersection

Q32 An ellipse centred at the origin has major axis along x, focus (3,0) and right directrix x=9. What are its eccentricity and full minor-axis length?
A) 1/3 and $6\sqrt{2}$
B) $1/\sqrt{3}$ and $6\sqrt{2}$
C) $1/\sqrt{3}$ and $3\sqrt{2}$
D) $\sqrt{3}$ and $6\sqrt{2}$

Q33 For which real a is $f(x)=x^3-3ax^2+3x$ strictly increasing for every real x?
A) $-1<a<1$
B) $a\leq -1$
C) $-1\leq a\leq 1$
D) $a\geq 1$

Q34 Find the total area between $y=x^2$ and $y=2-x^2$ over $-2\leq x\leq 2$.
A) 8
B) 4
C) 16/3
D) 32/3

Q35 Let $f(x)=x^2-4x+5$ with domain $x\geq 2$. Which inverse value $f^{-1}(10)$ is valid under that domain restriction?
A) Both -1 and 5
B) -1
C) 5
D) $\sqrt{10}$

Q36 For A with rows (1,2), (k,6), which k makes A singular?
A) 2
B) 6
C) 3
D) 12

Q37 Factory A supplies 2/3 of parts with 1% defective; B supplies 1/3 with 4% defective. Given a defective part, what is the chance it came from B?
A) 2/3
B) 1/3
C) 1/2
D) 4/5

Q38 Find the constant term in $(x^2-1/x)^6$ for $x\neq 0$.
A) -15
B) 15
C) 20
D) -20

Q39 An arithmetic progression has $S_5=25$ and $S_{10}=100$. What is its twentieth term?
A) 20
B) 37
C) 41
D) 39

Q40 For real a, $\lim(x\to0)[\sqrt{(1+ax)}-1]/x=3$. Find $\lim(x\to0)[\sqrt{(1+ax)}-1-3x]/x^2$.
A) 9/2
B) −3/2
C) 3/2
D) −9/2

Q41 L_1: r=(1,0,0)+t(1,1,0); L_2: r=(0,0,2)+s(0,1,1). Find their shortest distance.
A) 1
B) 2
C) √3
D) 1/√3

Q42 Six distinct people A–F sit at a round table, rotations identical but reflections distinct. How many arrangements have A not beside B and C not beside D?
A) 24
B) 48
C) 72
D) 96

Q43 For z=x+iy, impose |z−(2+i)|=|z−(−2+i)| and |z−i|=3. Which pair of z values satisfies both?
A) 4i and −2i
B) 3i and −3i
C) 2+4i and −2−2i
D) 4 and −2

Q44 Solve y′+2y=e^(−x), y(0)=0. What is y(ln 2)?
A) 1/2
B) 1/4
C) 3/4
D) 1

Q45 How many distinct solutions of sin(2x)=cos x lie in $0\le x<2\pi$?
A) 2
B) 3
C) 5
D) 4

Physics

Q46 Masses 3m and m hang over a light frictionless pulley fixed to a lift accelerating upward at g. Relative to the lift, what is the magnitude of their acceleration?
A) g
B) g/2
C) 2g
D) 0

Q47 A force in the direction of motion has magnitude F(x)=2x N when x is in metres. How much work does it do from x=0 to x=3 m?
A) 3 J
B) 6 J
C) 18 J
D) 9 J

Q48 Neglecting atmosphere and rotation, a planet's escape speed is v at radius R from its centre. What is escape speed at radius 4R?
A) v/4
B) 4v
C) v/2
D) 2v

Q49 A reversibly loaded material follows a straight stress–strain line from (0,0) to (2ε,2S), where ε and S are positive. What energy density is stored at strain 2ε?
A) Sε
B) 2Sε
C) 3Sε
D) 4Sε

Q50 An insulated vessel mixes 100 g ice at 0 °C with 300 g water at 50 °C. Use cwater=4.2 J g^{-1} K^{-1} and Lfusion=336 J g^{-1}. Neglect vessel heat capacity. Find final temperature.
A) 17.5 °C
B) 0 °C with ice remaining
C) 25 °C
D) 31.5 °C

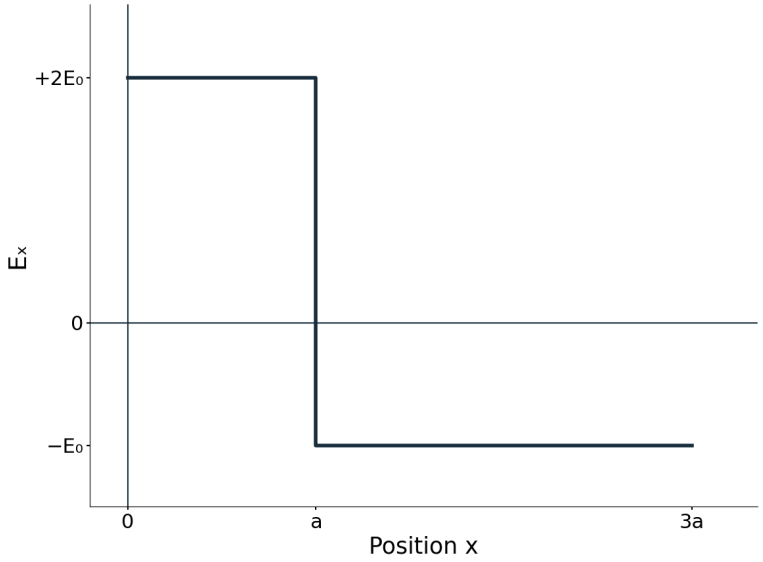
Q51 Two ideal gases at equal absolute temperature have molar masses in ratio 4:9. What is the ratio of their rms speeds in the same order?
A) 4:9
B) 2:3
C) 9:4
D) 3:2

Q52 Two identical springs of constant k support the same mass m, first in series and then in parallel. What is Tseries/Tparallel for small oscillations?
A) 1/2
B) √2
C) 2
D) 4

Q53 Capacitors C and 2C are in series with charge Q after disconnection from a source. A dielectric doubles only the first capacitance with no leakage. Find final/initial ratios of total stored energy and voltage on the modified element.
A) 3/2 and 1/2
B) 2/3 and 1/2
C) 2/3 and 2
D) 1 and 1/2

Q54 A positive charge q moves at speed v along +x in field $B(\hat{j}+\hat{k})$, with B>0. Which magnetic-force vector acts on it?
A) $qvB(\hat{k}-\hat{j})$
B) $qvB(\hat{j}-\hat{k})$
C) $qvB(\hat{j}+\hat{k})$
D) Zero

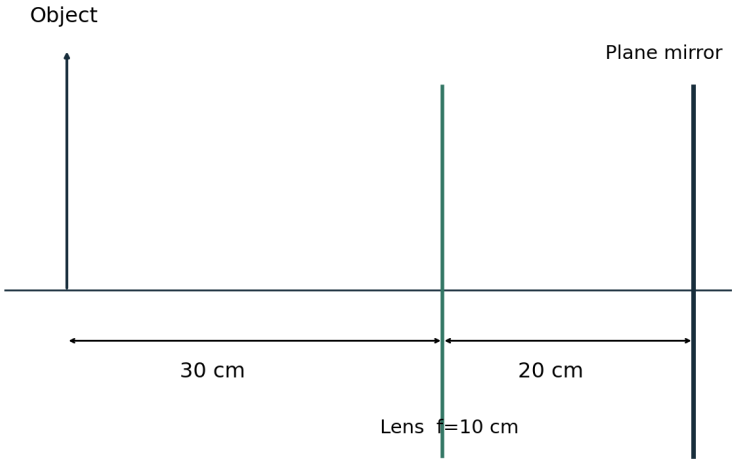
Q55 The figure shows E_x as a function of x between 0 and 3a, with E_0,a>0. What is V(3a)−V(0)?



A) $-4E_0a$
B) $-2E_0a$
C) $+2E_0a$
D) 0

Q56 A Wheatstone bridge has upper/lower left arms 2 Ω and 3 Ω and upper/lower right arms 4 Ω and R. An ideal galvanometer joins the midpoints. For zero galvanometer current, find R.
A) 2 Ω
B) 3 Ω
C) 6 Ω
D) 8 Ω

Q57 The figure places an object 30 cm left of a converging lens of focal length 10 cm and a plane mirror 20 cm right of the lens. After reflection and a second lens passage, where is the final image? Use paraxial rays.



A) 15 cm left of the lens
B) 50/3 cm left of the lens
C) 25 cm right of the lens
D) 50/3 cm right of the lens

Q58 Stopping potentials for light frequencies f and 2f are 1 V and 3 V for one metal. What is its threshold frequency in terms of f?
A) f/2
B) f/3
C) f
D) 2f

Q59 In hydrogen, what is wavelength ratio $\lambda(3\to2)/\lambda(2\to1)$ for emitted photons? Use $E_n=-13.6/n^2$ eV.
A) 5/27
B) 9/5
C) 5/9
D) 27/5

Q60 A loop lies in the page. Magnetic flux into the page increases. Viewed facing the page, what is the induced conventional current direction?
A) Clockwise, reinforcing inward flux
B) Clockwise, producing outward field
C) Counterclockwise, producing outward field
D) No current because field direction is unchanged

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A Seed-plant reproductive structures

Answer: (A) Gymnosperms.

Why: An exposed ovule can mature into a naked seed. Gymnosperms bear seeds without the enclosing ovary that later becomes a fruit; the two observations therefore reinforce the same identification. The ovule is borne directly on a cone scale rather than inside a carpel. Because no ovary wall develops around it, the mature seed is not enclosed by a fruit.

Why not the others: (B) Angiosperm ovules are inside an ovary and their seeds are enclosed in fruit. (C) Bryophytes reproduce by spores rather than seeds. (D) Pteridophytes also lack seeds and produce spores.

Key point: Locate the ovule before fertilisation and the seed after it.

Q2 / B Ovary position and floral whorls

Answer: (B) Inferior ovary; epigynous.

Why: Fusion of the floral tube around the ovary puts the apparent insertion of the other whorls above it. That relative position defines an inferior ovary and an epigynous flower, regardless of the flower's size. Use the level at which all other whorls arise relative to the ovary, not the flower's orientation on the page. The fused floral tube surrounds the ovary and makes it inferior.

Why not the others: (A) Hypogynous whorls arise below a superior ovary. (C) A perigynous cup surrounds but does not fuse to a superior ovary in this way. (D) Combines incompatible relative positions.

Key point: Judge ovary position from the insertion and fusion of floral parts.

Q3 / C Organelle identification by fractionation

Answer: (C) Fraction R.

Why: The figure shows strong catalase enrichment specifically in R, with little chlorophyll or ribosomal RNA. Catalase supports a peroxisome-rich preparation, while the low negative markers argue against chloroplast or ribosome contamination. Repeated isolation reduces the chance of a single separation accident. Catalase is a marker for peroxisomes because these organelles detoxify peroxide generated by oxidative metabolism. The other two marker profiles exclude a chloroplast- or ribosome-rich fraction even if isolation is imperfect.

Why not the others: (A) P's chlorophyll instead points to plastid material. (B) Q's RNA points to ribosome-rich material. (D) Protein occurs in many compartments and cannot identify an organelle.

Key point: Combine a positive marker with contamination controls.

Q4 / D Interacting limits on photosynthesis

Answer: (D) Low CO_2 limits the light response sooner.

Why: At high light, the low-CO_2 trace is nearly flat while the high-CO_2 trace continues rising. Increasing light therefore adds little when carbon supply is limiting, but increases carbon fixation when CO_2 is less restrictive. The comparison holds temperature fixed. The two curves share the same light input; their separation therefore diagnoses the changed carbon supply. At low light both traces may be light-limited, so a CO_2 response emerges chiefly after irradiance is sufficient.

Why not the others: (A) Neither curve decreases over the shown interval. (B) The high-CO_2 curve responds strongly to light. (C) Temperature was controlled, so the difference cannot be assigned to it.

Key point: Interpret paired interventions rather than naming a limiting factor from one curve.

Q5 / A Cardiac output and stroke volume

Answer: (A) 60 mL.

Given: Initial output is 60×80=4800 mL/min.

Formula/Principle: Cardiac output = heart rate × stroke volume.

Step 1: New output=1.25×4800=6000 mL/min.

Step 2: New stroke volume=6000/100=60 mL per beat.

Why: The output increases less proportionally than the rate, so the volume per beat must fall in this stipulated measurement. Cardiac output is rate times stroke volume, $Q = HR \times SV$. Comparing the two states gives the new stroke volume rather than assuming exercise preserves it; the rate alone cannot establish output.

Why not the others: (B) assumes unchanged stroke volume; (C) treats the new rate as volume; (D) doubles the correct value.

Key point: Calculate output before solving for a changed component.

Q6 / B Meiosis-I nondisjunction and chromosome number

Answer: (B) 11 and 13.

Given: Normal gametes have n=6. The affected meiosis-I daughters receive both homologues or neither.

Formula/Principle: Normal meiosis II separates sisters without correcting the unequal homologues.

Step 1: The four sperm therefore have n+1=7 or n−1=5 chromosomes.

Step 2: Each normal egg contributes six, yielding 13 or 11.

Why: The error happened before the reductional division completed, so none of this meiocyte's products has the normal count. A meiosis-I nondisjunction sends both homologues to one daughter and neither to the other. Meiosis II then separates sister chromatids normally, producing two n+1 and two n−1 gametes from this meiocyte.

Why not the others: (A) ignores nondisjunction; (C) counts two extra or missing homologues per gamete; (D) wrongly leaves one daughter normal.

Key point: Track homologues and sister chromatids in their respective divisions.

Q7 / C Codominance versus blending

Answer: (C) Codominance.

Why: Both parental traits are visibly represented in the same F_1 rather than one masking the other. Detecting both pigment types supports simultaneous expression, not an intermediate pigment. Patchy expression is not an intermediate blended colour: the parental pigments remain distinguishable in different regions. The biochemical detection of both products supports simultaneous allele expression rather than dominance of one.

Why not the others: (A) Would hide white expression. (B) Predicts an intermediate blended phenotype rather than separate patches. (D) Nothing about the cross associates the trait with sex.

Key point: Distinguish coexistence of parental phenotypes from an intermediate one.

Q8 / D Primary and secondary immune responses

Answer: (D) Immunological memory.

Why: Initial exposure activates antigen-specific lymphocytes and leaves memory cells after the first response contracts. Those cells are already selected for the same antigen, so the second exposure has a shorter lag and larger response at equal dose. The first exposure creates selected lymphocyte clones and a memory pool. On re-exposure those clones expand and differentiate sooner, explaining the shorter lag and higher titre without requiring a larger antigen dose.

Why not the others: (A) Antigen recognition is still necessary. (B) Innate cells do not themselves turn into antibodies. (C) The experiment holds antigen identity fixed.

Key point: Compare response timing under a controlled repeat exposure.

Q9 / A Cohesive-end ligation

Answer: (A) Alignment by base pairing.

Why: Complementary short overhangs transiently anneal, increasing the chance that the vector and insert ends remain together for ligase to seal the backbone. This is a pairing advantage, not an absolute chemical requirement of ligase. Compatible sticky ends increase encounter specificity through base pairing but do not create a covalent phosphodiester backbone. DNA ligase seals the sugar–phosphate nicks, making a stable recombinant molecule.

Why not the others: (B) Blunt ends can also be ligated. (C) Identical compatible ends can permit either orientation. (D) Only the overhang region is single-stranded.

Key point: Separate cohesive-end annealing from phosphodiester bond formation.

Q10 / B Density-dependent population growth

Answer: (B) Resource-limited growth.

Why: Increments rise at first but fall sharply near the last counts, even though the population remains positive and growing. With fixed food and no migration, density-dependent limitation is a coherent explanation for approaching a carrying level. The counts alone do not establish its exact value. As population size approaches environmental carrying capacity, per-capita resource availability falls and net growth slows. A rising total count can coexist with shrinking increments, so absolute increase alone does not prove exponential growth.

Why not the others: (A) Fixed exponential growth would not flatten this way. (C) Every recorded count rises. (D) Migration was excluded.

Key point: Compare successive increments, not just population sizes.

Q11 / C Conditional recessive pedigree risk

Answer: (C) 1/12.

Given: The affected sibling establishes carrier parents in the first family.

Formula/Principle: Condition on the person's unaffected status, then multiply independent carrier and transmission probabilities.

Step 1: Among unaffected children of Aa×Aa, two of three genotype classes are carriers: 2/3.

Step 2: Partner carrier probability from Aa×AA is 1/2; two carriers have an affected child with probability 1/4. Product=(2/3)(1/2)(1/4)=1/12.

Why: The condition of being unaffected removes the aa child from the first denominator. The parents must both be carriers because an aa child appeared. Of the four equally likely offspring genotypes, only AA, Aa, Aa remain after conditioning on being unaffected, making two of three carriers.

Why not the others: (A) uses 1/3 carrier risk; (B) treats carrier status as certain; (D) omits the 1/4 child factor.

Key point: Recondition a pedigree probability when phenotype is known.

Q12 / B Reading-frame shift after insertion

Answer: (B) Downstream frame shift.

Why: The ribosome reads consecutive, non-overlapping groups of three bases from the same initiation point. Adding one base after AUG changes which bases form every subsequent triplet until translation stops; even the location of a stop codon can change. An insertion not divisible by three changes every downstream codon boundary until a stop is encountered. Compare the new triplet grouping from the insertion point; simply adding one amino acid would assume the original frame remains intact.

Why not the others: (A) Assumes the insertion replaced only one whole codon. (C) AUG lies before the insertion. (D) Triplet grouping cannot remain the same after a one-base insertion.

Key point: Track reading positions, not just the inserted letter.

Q13 / D Filtered and excreted solute loads

Answer: (D) 90%.

Given: Filtered load=100 mL/min×1 mg/mL=100 mg/min; excretion=1 mL/min×10 mg/mL=10 mg/min.

Formula/Principle: Reabsorbed load=filtered load−excreted load when secretion is absent.

Step 1: Reabsorbed=90 mg/min.

Step 2: Fraction=90/100=0.90.

Why: Concentrations alone cannot be compared because plasma filtration and urine flows differ. Filtered load is $P \times GFR$ in mass per minute, whereas excreted load is $U \times \dot{V}$. Only after converting both to the same units can net reabsorption or secretion be inferred.

Why not the others: (A) confuses urine flow with fraction; (B) is excreted fraction; (C) ignores loads; none gives 90%.

Key point: Convert concentrations to rates before subtracting.

Q14 / C Spindle checkpoint and anaphase

Answer: (C) Chromatid separation.

Why: Unattached kinetochores fail the spindle-attachment checkpoint, preventing the coordinated onset of anaphase. Chromosomes can condense, yet sisters cannot segregate reliably to opposite poles. DNA replication has already occurred under the stated condition. Kinetochores monitor bipolar attachment before anaphase. Although chromatids have already been copied, the checkpoint inhibits separation when spindle capture is incomplete, preventing unequal segregation.

Why not the others: (A) Is expressly unaffected. (B) Condensed mitotic cells have reached a later stage. (D) Protein synthesis is not the specified inhibited event.

Key point: An attachment checkpoint links spindle status to chromatid separation.

Q15 / A Rubisco oxygenation and photorespiration

Answer: (A) More photorespiration.

Why: Rubisco can catalyse both carboxylation and oxygenation. Raising O_2 relative to CO_2 shifts competition toward oxygenation, initiating photorespiratory reactions that consume resources and reduce net carbon gain under the controlled conditions. It need not stop all carboxylation. Oxygen competes with carbon dioxide for Rubisco's active site. Oxygenation initiates photorespiration, consuming energy and releasing previously fixed carbon, thereby lowering net fixation despite equal stomatal conductance.

Why not the others: (B) Ignores Rubisco's oxygenase activity. (C) O_2 supplies no carbon for fixation. (D) Reduced net gain is not complete abolition of the cycle.

Key point: Compare competing substrates at the same enzyme.

Chemistry

Q16 / D Lone-pair repulsion and bond angle

Answer: (D) Lone-pair–bond-pair repulsion.

Why: Both centres have four electron domains, but NH_3 has three bond pairs and a lone pair. The more spatially demanding lone pair compresses the angles between N–H bonds below the ideal tetrahedral angle of CH_4. A lone pair occupies more angular space around nitrogen than a shared bond pair, compressing the three N–H directions. Carbon's four equivalent bonding domains avoid this extra compression.

Why not the others: (A) NH_3 has three N–H bonds. (B) Carbon has four bonds and no lone pair here. (C) Electron-domain geometry is approximately tetrahedral even though molecular geometry is pyramidal.

Key point: Distinguish electron-domain and molecular geometry.

Q17 / A Ionisation isomerism and counterion tests

Answer: (A) P gives $BaSO_4$, Q gives AgBr.

Why: Brackets in the figure delimit each coordination sphere. P has freely dissociated sulfate that Ba^{2+} removes as insoluble $BaSO_4$; Q has free bromide that Ag^+ removes as AgBr. The coordinated ligands are not assumed to leave immediately. The free counter-ion can react immediately in solution, while a ligand inside the square-bracket coordination sphere does not behave as the same free ion. Read each bracket before predicting the precipitation test.

Why not the others: (B) Reverses inner and outer sphere. (C) Treats every coordinated ligand as a counterion. (D) Ignores the actual counterions.

Key point: Read the brackets before predicting precipitation.

Q18 / B Hydrogen ionisation from an excited level

Answer: (B) 3.4 eV.

Given: $E_2 = -13.6/4 = -3.4$ eV and $E_\infty = 0$.

Formula/Principle: Absorption needs the energy difference final minus initial.

Step 1: $\Delta E = 0 - (-3.4) = +3.4$ eV.

Worked derivation:

$$E_2 = -13.6/2^2 = -3.40 \text{ eV},$$
$$E_{\text{ionise}} = E_\infty - E_2 = 0 - (-3.40) = +3.40 \text{ eV}.$$

Why: The electron starts already above the ground state, so less than 13.6 eV is needed. Ionisation from $n = 2$ means lifting the electron from $-13.6/4 = -3.4$ eV to the continuum at zero. The photon must supply that energy gap, not the ground-state binding energy.

Why not the others: (A) Ground-state ionisation; (C) n=1→2 excitation; (D) Confuses the n=3 binding energy.

Key point: Set the continuum energy to zero before subtracting.

Q19 / C Carbocation rearrangement

Answer: (C) A more stable carbocation.

Why: During a hydride shift, the C–H bonding pair migrates to the electron-deficient adjacent carbon. Positive charge then resides at the carbon that lost hydride, which is more substituted under the stated structure and stabilised by alkyl donation. The hydride moves with two bonding electrons; the formally positive site appears on the carbon it leaves. Compare the substitution of the old and new cation centres to decide whether rearrangement is stabilising.

Why not the others: (A) No ring is specified. (B) Charge remains positive. (D) A hydride shift does not cleave all C–C bonds.

Key point: Track both the migrating electron pair and the new charge site.

Q20 / D Peroxide effect in HBr addition

Answer: (D) 2-bromopropane then 1-bromopropane.

Why: Ionic addition without peroxide proceeds through the more stable secondary carbocation, placing bromine on the middle carbon. The peroxide-initiated HBr radical chain favours the orientation that forms the more stable carbon radical and ultimately puts bromine on the terminal carbon. Peroxide supplies a radical chain pathway for HBr in the stated conditions; the Br radical adds so the more stable carbon radical forms before H abstraction. The ionic route instead optimises carbocation stability.

Why not the others: (A) Reverses the conditions. (B) Applies the radical result to both. (C) Omits the peroxide effect.

Key point: Choose the mechanism before choosing regioselectivity.

Q21 / A Substitution versus elimination pathways

Answer: (A) Inversion and a more substituted trans-favoured alkene.

Why: Backside displacement reverses the arrangement at the reacting stereocentre. E2 removes a β hydrogen with departure of bromide; with a small base, but-2-ene is generally favoured over but-1-ene, and its trans form is more stable than cis. These are separate stereochemical and regioselective decisions. An SN2 transition state requires backside approach and gives inversion at the stereocentre. E2 forms a double bond by removing a β proton and leaving group together, so the product lacks the same tetrahedral stereocentre.

Why not the others: (B) Gives retention and the less substituted alkene. (C) Assigns planar-intermediate racemisation to SN2. (D) Names a substitution alcohol rather than an E2 product.

Key point: Analyse each reaction condition independently.

Q22 / B Phenol-specific ferric chloride test

Answer: (B) X phenol, Y alcohol.

Why: Sodium reacts with acidic O–H groups in both alcohol and phenol, so that test alone does not distinguish them. A neutral ferric chloride colour is characteristic of the phenolic group in this qualitative comparison, selecting X as phenol. Neutral ferric chloride responds to phenolic oxygen through a coloured complex; sodium alone tests for sufficiently acidic O–H and is not selective. Combining both tests distinguishes phenol from a simple alcohol.

Why not the others: (A) Reverses the distinguishing observation. (C) Ether and alkane lack the O–H needed for the stated sodium response. (D) An alkane cannot explain X's tests.

Key point: Interpret the two tests together, not independently.

Q23 / C Aldol condensation of ethanal

Answer: (C) 3-hydroxybutanal then but-2-enal.

Why: Ethanal's enolate attacks the carbonyl carbon of a second ethanal, joining two C_2 units. Protonation yields CH_3–$CH(OH)$–CH_2–CHO, with the OH at the β carbon. Subsequent dehydration removes water across the α–β positions to give CH_3–$CH{=}CH$–CHO. The first low-temperature step creates a β-hydroxy aldehyde, not an α,β-unsaturated product. Warming removes water across the adjacent carbons, introducing the conjugated C=C while retaining the carbonyl group.

Why not the others: (A) Places OH at the wrong carbon and reduces rather than dehydrates. (B) Loses one carbon from the two ethanal units. (D) Places the carbonyl internally and loses oxygen entirely.

Key point: Count carbons and track the β-hydroxy carbonyl before eliminating water.

Q24 / D Aromatic versus aliphatic amine basicity

Answer: (D) Resonance lowers lone-pair availability.

Why: The aniline lone pair overlaps with the aromatic π system, stabilising the unprotonated base. Cyclohexylamine lacks this conjugation, so its lone pair is more available to accept a proton under comparable aqueous conditions. Solvation can matter, but the proposed resonance distinction is the central structural reason. Resonance donation of the aniline lone pair into the ring lowers its availability to bind a proton. The cyclohexylamine lone pair is localised on nitrogen and the alkyl framework does not delocalise it.

Why not the others: (A) The lone pair exists. (B) Cyclohexylamine is an amine. (C) Both nitrogens have ordinary amine valence.

Key point: Compare conjugation of the free base when assessing basicity.

Q25 / A Osmotic pressure of mixed nonelectrolytes

Answer: (A) 2.5.

Given: Total dissolved particle concentration is (0.020+0.030)/1.00=0.050 M.

Formula/Principle: Ideal osmotic pressure π=CRT for nonelectrolytes.

Step 1: Temperature and R cancel in the ratio.

Step 2: πmixture/πglucose=0.050/0.020=2.5.

Worked derivation:

$$\begin{aligned} C_{\text{mix}} &= (0.020 + 0.030)/1.00 = 0.050\ \text{M}, \\ C_{\text{glucose}} &= 0.020\ \text{M}, \\ \Pi_{\text{mix}}/\Pi_{\text{glucose}} &= 0.050RT/(0.020RT) = 2.5. \end{aligned}$$

Why: Each dissolved molecule contributes a particle; no ionisation factor is needed. Both glucose and urea remain as molecules, so total solute concentration is $0.020 + 0.030 = 0.050\,\text{mol L}^{-1}$. Osmotic pressure uses that total particle molarity, not a separate ionic multiplier.

Why not the others: (B) Counts only urea relative to glucose; (C) reverses the ratio; (D) adds an unjustified dissociation factor.

Key point: Sum particle molarities, not solute identities.

Q26 / B Silver concentration-cell polarity and emf

Answer: (B) The dilute electrode; +0.118 V.

Given: Silver reduction potential increases with [Ag^+].

Formula/Principle: E=E°+0.059 log[Ag^+] for $Ag^+ + e^- \rightarrow Ag$; Ecell=Ecathode−Eanode.

Step 1: Concentrated side is the cathode; dilute side oxidises and is negative.

Step 2: Ecell=0.059 log(1.0/0.010)=0.118 V.

Worked derivation:

$$\begin{aligned} \text{Ag}^+ + \text{e}^- &\longrightarrow \text{Ag}, \\ E_{\text{cell}} &= 0.059\,\log_{10}(1.0/0.010) \\ &= 0.059(2) = 0.118\ \text{V}. \end{aligned}$$

Why: Oxidation on the dilute side tends to replenish Ag^+ and diminish the concentration difference. At the dilute electrode oxidation generates Ag^+, while reduction at the concentrated electrode removes it. This direction reduces the concentration gradient and is consistent with a positive spontaneous cell emf.

Why not the others: (A) Reverses polarity; (C) counts only one decade; (D) reverses polarity and emf sign.

Key point: Determine electrode roles before evaluating logarithms.

Q27 / C Orders from controlled initial-rate experiments

Answer: (C) k[A][B]².

Given: Trial 1→2 doubles A alone and rate 2→4; trial 2→3 doubles B alone and rate 4→16.

Formula/Principle: At fixed other reactants, a doubling changes rate by 2 raised to that reactant's order.

Step 1: A factor 2 implies first order in A.

Step 2: B factor 4 implies second order in B.

Worked derivation:

$$2^m = 4/2 = 2 \Rightarrow m = 1,$$
$$2^n = 16/4 = 4 \Rightarrow n = 2,$$
$$v = k[A][B]^2.$$

Why: The controlled comparisons isolate one variable at a time, so a combined trial 1→3 ratio would obscure the separate exponents. Trial 1→2 changes only A, so its rate ratio determines A's order; trial 2→3 changes only B. Reading the bar heights and concentration labels together prevents attribution of a combined change to one reactant.

Why not the others: (A) swaps orders; (B) underpredicts trial 3; (D) overpredicts trial 2.

Key point: Read the trial labels as well as bar heights.

Q28 / D Hess law and reaction reversal

Answer: (D) +35 kJ/mol.

Given: A→C is the sum of the two given paths.

Formula/Principle: Enthalpy is a state function; reversing a reaction reverses ΔH.

Step 1: ΔH(A→C)=40−75=−35 kJ/mol.

Step 2: Therefore C→A has +35 kJ/mol.

Worked derivation:

$$\Delta H_{A\to C} = 40 - 75 = -35, \quad \Delta H_{C\to A} = -\Delta H_{A\to C} = +35 \text{ kJ mol}^{-1}.$$

Why: Endpoint states, not the route, fix the net enthalpy change. Hess's law gives $\Delta H_{A\to C} = 40 - 75 = -35\,\text{kJ mol}^{-1}$, so reversing the endpoints changes the sign. An intermediate path cannot change the state-function difference.

Why not the others: (A) adds magnitudes then reverses; (B) forgets reversal; (C) adds magnitudes with wrong sign.

Key point: Draw the directed path before combining changes.

Q29 / A Strong acid-base titration beyond equivalence

Answer: (A) pH 12.30.

Given: Acid=2.0 mmol; added base=3.0 mmol; final volume=50.0 mL.

Formula/Principle: Neutralise first; then pOH=−log[OH⁻], pH=14−pOH.

Step 1: Excess OH⁻=1.0 mmol/50.0 mL=0.020 M.

Step 2: pOH≈1.70 and pH≈12.30.

Worked derivation:

$$n_{\text{OH,excess}} = (0.0300 - 0.0200)(0.10) = 0.00100 \text{ mol},$$
$$\left[\text{OH}^-\right] = 0.00100/0.0500 = 0.0200 \text{ M},$$
$$pH = 14 + \log_{10}(0.0200) \approx 12.30.$$

Why: Past equivalence, excess strong base controls pH rather than the neutral salt. The extra hydroxide is the initial acid-neutralisation deficit in reverse: calculate remaining OH⁻ moles after reaction, divide by the *combined* volume, then convert pOH to pH at the stated temperature.

Why not the others: (B) Reports pOH; (C) assumes equivalence; (D) neglects excess-base dilution.

Key point: Count moles before taking a logarithm.

Q30 / B Spin-only magnetic moment

Answer: (B) √3 BM.

Given: Number of unpaired electrons n=1.

Formula/Principle: Spin-only μ=√[n(n+2)] BM.

Step 1: Substitute n=1: μ=√(1×3)=√3 BM.

Worked derivation:

$$\mu = \sqrt{n(n+2)}\,\mu_B = \sqrt{1(3)}\,\mu_B = \sqrt{3}\,\mu_B.$$

Why: Spin-only counting depends on unpaired spins, not the total electron number or metal identity under this stipulated model. The spin-only relation is $\mu = \sqrt{n(n+2)}$ BM. With $n = 1$, the moment follows from one unpaired spin; ligand and orbital contributions are excluded by the stated approximation.

Why not the others: (A) Omits the n+2 factor; (C) corresponds to n=2; (D) drops the square root.

Key point: Count unpaired electrons before evaluating the moment.

Mathematics

Q31 / C Distance from a centre to a tangent

Answer: (C) 4±√7.

Given: Tangency to x-axis fixes radius 4.

Formula/Principle: Circle equation (x−3)²+(y−4)²=16.

Step 1: On the y-axis x=0, so (y−4)²=7.

Step 2: y=4±√7, and both values are positive.

Worked derivation:

$$(x-3)^2 + (y-4)^2 = 4^2, \quad x = 0 \Longrightarrow (y-4)^2 = 7.$$

Why: A vertical line through the y-axis meets this circle twice because the centre is only three units from it. The positive-y-axis condition must be checked separately for both algebraic roots: since √7<4, neither root lies below the origin.

Why not the others: (A) and (B) each omit one intersection; (D) incorrectly compares radius with the centre's horizontal distance.

Key point: Check both square-root branches against the stated ray.

Q32 / B Ellipse focus and directrix

Answer: (B) 1/√3 and 6√2.

Given: c=3 and right directrix a/e=9.

Formula/Principle: c=ae, b²=a²−c².

Step 1: a/e=a²/c=9, so a²=27 and e=3/√27=1/√3.

Step 2: b²=27−9=18; full minor axis 2b=6√2.

Worked derivation:

$$e = \frac{c}{a} = \frac{a}{9}, \quad c = 3 \Longrightarrow a^2 = 27; \qquad b^2 = a^2 - c^2 = 18.$$

Why: Both focus and directrix fix a rather than either datum alone. The focus and directrix jointly fix eccentricity and semimajor axis. The minor axis is twice b, whereas a positive eccentricity is less than one.

Why not the others: (A) Uses 3/9 for e; (C) halves the axis length; (D) has e>1 and is not an ellipse.

Key point: Distinguish semiaxis b from axis length 2b.

Q33 / C Global monotonicity with parameter endpoints

Answer: (C) $-1 \le a \le 1$.

Given: $f'(x)=3[(x-a)^2+1-a^2]$.

Formula/Principle: A derivative nonnegative everywhere with only isolated zeros yields a strictly increasing polynomial.

Step 1: The minimum derivative is $3(1-a^2)$, requiring $|a| \le 1$.

Step 2: At ± 1 it vanishes only at one point, so no interval is constant and endpoints remain valid.

Worked derivation:

$$f'(x) = 3(x-a)^2 + 3(1-a^2), \qquad \min_{x \in \mathbb{R}} f'(x) = 3(1-a^2).$$

Why: Requiring $f'>0$ everywhere would incorrectly discard boundary cases. At $|a|=1$ the derivative is zero at a single x but positive everywhere else; integrating it over any nonzero interval still gives a positive change in f.

Why not the others: (A) Excludes valid endpoints; (B) and (D) include values beyond the permitted range.

Key point: Check whether isolated stationary points spoil strict monotonicity.

Q34 / A Unsigned area across curve crossings

Answer: (A) 8.

Given: Curves cross at $x=\pm 1$, so the upper curve changes.

Formula/Principle: Area$=\int_{-2}^{2}|2-2x^2|dx$, not a signed net integral.

Step 1: By even symmetry, area$=2[\int_0^1(2-2x^2)dx+\int_1^2(2x^2-2)dx]$.

Step 2: The two bracketed integrals are 4/3 and 8/3, giving $2\times 4=8$.

Worked derivation:

$$x^2 = 2 - x^2 \Longrightarrow x = \pm 1; \quad A = \int_{-2}^{-1} (2x^2-2)\,dx + \int_{-1}^{1} (2-2x^2)\,dx + \int_{1}^{2} (2x^2-2)\,dx = 8.$$

Why: The outer lobes must be added after reversing their signed integrand. Area uses absolute vertical separation. The sign of the difference reverses at both intersections, so a single signed integral across the full interval would cancel regions.

Why not the others: (B) drops the symmetry factor; (C) counts only outer lobes; (D) results from using one sign over all intervals.

Key point: Split at every crossing when asked for area.

Q35 / C Inverse on a restricted domain

Answer: (C) 5.

Given: $f(x)=(x-2)^2+1$ and its domain is $x \ge 2$.

Formula/Principle: Inverse evaluation means solving $f(x)=10$ and retaining only inputs in the given domain.

Step 1: $(x-2)^2=9$, so $x=2\pm 3$, i.e. 5 or -1.

Step 2: Reject $-1<2$; $f^{-1}(10)=5$.

Worked derivation:

$$f(x) = (x-2)^2 + 1, \quad x \ge 2; \qquad f^{-1}(u) = 2 + \sqrt{u-1}, \quad f^{-1}(10) = 5.$$

Why: The restricted parabola is one-to-one, whereas the unrestricted parabola is not. The domain restriction selects the positive square-root branch. The algebraic alternative x=-1 gives the same function value but is outside the specified domain.

Why not the others: (A) retains the prohibited branch; (B) selects that branch alone; (D) omits the translation and square-root relation.

Key point: Apply the domain restriction after solving the inverse equation.

Q36 / C Singularity from dependent rows

Answer: (C) 3.

Given: A is a 2×2 matrix.

Formula/Principle: A square matrix is singular precisely when its determinant vanishes.

Step 1: det A=1×6−2k=6−2k.

Step 2: Setting it to zero gives k=3.

Worked derivation:

$$\det A = (1)(6) - (2)k = 6 - 2k; \qquad \det A = 0 \Longrightarrow k = 3.$$

Why: At this value the second row is three times the first, so the columns cannot be independent. A zero determinant is equivalent to row dependence here: at k=3 the second row is three times the first, so the matrix cannot have an inverse.

Why not the others: (A), (B) and (D) give determinants 2, −6 and −18, respectively, all nonzero.

Key point: Confirm a zero determinant by row dependence.

Q37 / A Unequal priors and conditional probability

Answer: (A) 2/3.

Given: Defective shares are (2/3)(.01) from A and (1/3)(.04) from B.

Formula/Principle: Posterior share=source's joint defective probability divided by all defective probability.

Step 1: Contributions have ratio 2:4.

Step 2: B share=4/(2+4)=2/3.

Worked derivation:

$$P(B \mid D) = \frac{P(B)P(D \mid B)}{P(A)P(D \mid A) + P(B)P(D \mid B)} = \frac{(1/3)(.04)}{(2/3)(.01) + (1/3)(.04)} = \frac{2}{3}.$$

Why: A's larger output does not cancel B's larger defect rate. A defect changes the source probabilities because B produces defects four times as often. Both prior production shares must remain in the denominator.

Why not the others: (B) uses the prior alone; (C) assumes equal contributions; (D) ignores source proportions.

Key point: Weight likelihoods by source frequency.

Q38 / B Constant term with signed exponents

Answer: (B) 15.

Given: Choosing k factors −1/x leaves x exponent 2(6−k)−k=12−3k.

Formula/Principle: Binomial term is C(6,k)(x²)^(6−k)(−1/x)^k.

Step 1: Set exponent zero: k=4.

Step 2: Coefficient=C(6,4)(−1)^4=15.

Worked derivation:

$$T_k = \binom{6}{k}(-1)^k x^{12-3k}; \quad 12-3k=0 \Longrightarrow k=4, \qquad T_4 = \binom{6}{4} = 15.$$

Why: The even selected index makes the sign positive. The exponent criterion selects one term, then its binomial sign is evaluated separately; substituting k before checking parity risks a negative distractor.

Why not the others: (A) loses parity; (C) uses k=3; (D) makes both mistakes.

Key point: Solve the exponent condition before the coefficient.

Q39 / D Arithmetic progression from two partial sums

Answer: (D) 39.

Given: S_n=n[2a+(n−1)d]/2.

Formula/Principle: Two sums determine initial term a and common difference d.

Step 1: S_5=25 gives a+2d=5; S_{10}=100 gives 2a+9d=20.

Step 2: Subtract twice the first relation: 5d=10, so d=2,a=1. Then a_{20}=a+19d=39.

Worked derivation:

$$S_n = \frac{n}{2}[2a+(n-1)d]; \quad 2a+4d=10, \quad 2a+9d=20 \Longrightarrow d=2,\ a=1,\ a_{20}=1+19(2)=39.$$

Why: Using S_{10}−S_5 gives a five-term sum, not the twentieth term. The two partial sums yield independent linear constraints on first term and common difference; one sum alone cannot determine both.

Why not the others: (A) confuses n and term; (B) is a_{19}; (C) is a_{21}.

Key point: Use sums to recover the underlying sequence.

Q40 / D Second-order radical limit from a first-order condition

Answer: (D) −9/2.

Given: First limit equals a/2, hence a=6.

Formula/Principle: Expansion √(1+t)=1+t/2−t²/8+o(t²) near zero.

Step 1: √(1+6x)=1+3x−36x²/8+o(x²).

Step 2: Divide the residual by x² to get −9/2.

Worked derivation:

$$\sqrt{1+ax} = 1+\frac{a}{2}x-\frac{a^2}{8}x^2+O(x^3), \quad a/2=3 \Longrightarrow a=6, \quad -a^2/8=-9/2.$$

Why: The subtraction removes the first-order term; retaining only that term yields a false zero. The first limit fixes the linear coefficient. Subtracting that linear contribution exposes the quadratic coefficient, so direct substitution before cancellation gives an indeterminate form.

Why not the others: (A) loses the negative sign; (B) and (C) omit the square of a.

Key point: Use the first limit to fix the parameter before the second-order limit.

Q41 / D Distance between skew lines

Answer: (D) 1/√3.

Given: Direction vectors u=(1,1,0), v=(0,1,1); displacement from L_1 to L_2 is w=(−1,0,2).

Formula/Principle: For nonparallel skew lines, distance=|w·(u×v)|/|u×v|.

Step 1: u×v=(1,−1,1), magnitude √3.

Step 2: w·(u×v)=−1+2=1; distance=1/√3.

Worked derivation:

$$d = \frac{|(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u}_1 \times \mathbf{u}_2)|}{|\mathbf{u}_1 \times \mathbf{u}_2|}, \quad \mathbf{u}_1 \times \mathbf{u}_2 = (1,-1,1), \quad d = \frac{1}{\sqrt{3}}.$$

Why: Only the component of separation along the common normal cannot be removed by moving along either line. The cross product is normal to both line directions; the scalar triple product projects their separation onto this common normal, eliminating shifts along either line.

Why not the others: (A) omits normalisation; (B) uses vertical displacement; (C) multiplies instead of dividing by √3.

Key point: Project separation onto the unit common normal.

Q42 / B Two forbidden circular adjacencies

Answer: (B) 48.

Given: Total circular arrangements (6−1)!=120.

Formula/Principle: Inclusion-exclusion over adjacency AB and CD.

Step 1: Each adjacent pair as a block gives 2(5−1)!=48 arrangements.

Step 2: Both pairs as blocks give 2^2(4−1)!=24.

Step 3: Neither=120−48−48+24=48.

Worked derivation:

$$\text{all circular orders} = (6-1)! = 120; \quad \text{adjacent } A,B = 2(5-1)! = 48.$$

Why: The overlap must be restored after subtracting the two forbidden sets. Treat A and B as one block, then count the two possible internal orders. Rotations are identified only after the block is formed; mirror images remain distinct.

Why not the others: (A) counts the overlap; (C) subtracts only one event; (D) omits the overlap correction.

Key point: Count rotations once at every block stage.

Q43 / A Intersection of complex loci

Answer: (A) 4i and −2i.

Given: Equal distance from (2,1) and (−2,1) puts the point on their perpendicular bisector.

Formula/Principle: First locus x=0; second circle $x^2+(y-1)^2=9$.

Step 1: Substitute x=0 to obtain y−1=±3.

Step 2: y=4 or −2, hence z=4i or −2i.

Worked derivation:

$$|z-(2+i)|^2 = |z-(-2+i)|^2 \Longrightarrow x = 0; \quad |iy - i| = 3 \Longrightarrow |y-1| = 3,\ y = 4, -2.$$

Why: Both constraints must hold; the first locus alone is a whole line. The first locus is the perpendicular bisector of two horizontally separated points, not a circle. Intersecting that line with the second circle gives two solutions.

Why not the others: (B) uses circle centred at origin; (C) violates x=0; (D) treats the imaginary coordinate as real.

Key point: Intersect loci geometrically before solving algebraically.

Q44 / B First-order linear equation with initial value

Answer: (B) 1/4.

Given: Integrating factor e^(2x).

Formula/Principle: (e^(2x)y)′=e^x.

Step 1: Integrate: e^(2x)y=e^x+C; y(0)=0 gives C=−1.

Step 2: y=e^(−x)−e^(−2x); at ln 2, y=1/2−1/4=1/4.

Worked derivation:

$$\frac{d}{dx}(e^{2x}y) = e^x, \quad e^{2x}y = e^x - 1, \quad y(\ln 2) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Why: The integrating factor converts the linked y′ and y terms into one product derivative. Multiplication by the integrating factor changes the left side into an exact product derivative. The initial condition is applied to the integrated expression before evaluating the requested point.

Why not the others: (A) drops the homogeneous term; (C) adds instead of subtracts; (D) ignores the initial condition.

Key point: Check the constant with the initial value before evaluation.

Q45 / D Root counting after factorisation

Answer: (D) 4.

Given: sin2x=2sin x cos x.

Formula/Principle: Factor the equation as cos x(2sin x−1)=0 without dividing by cos x.

Step 1: cos x=0 gives π/2 and 3π/2.

Step 2: sin x=1/2 gives π/6 and 5π/6, all in the half-open interval.

Worked derivation:

$$\sin 2x = \cos x \iff \cos x(2\sin x - 1) = 0; \quad x \in \{\pi/2, 3\pi/2, \pi/6, 5\pi/6\}.$$

Why: Dividing by cos x would lose two valid solutions. Factoring rather than dividing by cos x preserves two legitimate roots. The half-open interval includes zero but excludes 2π; neither endpoint adds a solution.

Why not the others: (A) counts only one factor; (B) misses a quadrant; (C) double-counts a boundary or extraneous point.

Key point: Preserve zero-factor branches before simplifying.

Physics

Q46 / A Atwood system inside accelerating elevator

Answer: (A) g.

Given: The lift's effective downward gravity is g+g=2g.

Formula/Principle: Atwood acceleration=(M−m)/(M+m) times effective gravity.

Step 1: Mass ratio=(3m−m)/(3m+m)=1/2.

Step 2: Relative acceleration=(1/2)(2g)=g.

Worked derivation:

$$g_{\text{eff}} = g + g = 2g, \qquad a = \frac{3m - m}{3m + m}\, g_{\text{eff}} = \frac{2}{4}(2g) = g.$$

Why: In the noninertial lift frame, the upward lift acceleration adds a downward pseudo acceleration. The lift frame adds a downward fictitious acceleration, so the Atwood relation uses 2g rather

than g. Both masses share the same acceleration magnitude along the taut string.

Why not the others: (B) ignores lift motion; (C) ignores mass ratio; (D) incorrectly equates unequal tensions and weights.

Key point: Specify the frame before using an effective g.

Q47 / D Work by a position-dependent force

Answer: (D) 9 J.

Given: Force rises linearly from 0 to 6 N.

Formula/Principle: Work=∫F dx, the area under the force–displacement curve.

Step 1: $\int_0^3 2x\,dx=[x^2]_0^3=9$ N·m.

Step 2: Convert N·m to 9 J.

Worked derivation:

$$W=\int_0^{3\,\mathrm{m}} F(x)\,dx=\int_0^3 2x\,dx=[x^2]_0^3=9\,\mathrm{J}.$$

Why: Using the final force for the whole interval overcounts the work because the force was smaller earlier. Work is area under the force–displacement curve. Multiplying the endpoint force by total distance would overcount because the force grows from zero.

Why not the others: (A) drops the coefficient; (B) uses an incorrect average force; (C) uses final force throughout.

Key point: Integrate when force changes with position.

Q48 / C Escape speed versus distance from centre

Answer: (C) v/2.

Given: The same planet has mass M at both radii.

Formula/Principle: vesc=√(2GM/r).

Step 1: Ratio vesc(4R)/vesc(R)=√(R/4R).

Step 2: This is 1/2.

Worked derivation:

$$\frac{v_{\mathrm{esc}}(4R)}{v_{\mathrm{esc}}(R)}=\frac{\sqrt{2GM/(4R)}}{\sqrt{2GM/R}}=\sqrt{\frac{1}{4}}=\frac{1}{2}.$$

Why: Escape energy depends on gravitational potential GM/r, and speed is its square root. The planet's mass and gravitational constant cancel in a ratio. The same formula assumes the point is outside the planet and neglects other bodies.

Why not the others: (A) scales speed linearly with 1/r; (B) and (D) make speed rise with distance.

Key point: Distinguish inverse-radius energy from inverse-square-root speed.

Q49 / B Strain energy as stress–strain area

Answer: (B) 2Sε.

Given: Stress increases linearly to 2S over strain range 2ε.

Formula/Principle: Reversible strain energy per volume is ∫stress d(strain), the triangular area.

Step 1: Area=(1/2)(2ε)(2S).

Step 2: The result is 2Sε, with pressure units J/m³.

Worked derivation:

$$u=\int_0^{2\varepsilon}\sigma(\varepsilon')\,d\varepsilon'=\tfrac{1}{2}(2\varepsilon)(2S)=2S\varepsilon.$$

Why: Each small strain increment stores its current stress times the increment, not the final stress for the entire path. For a reversible straight loading line, the recoverable energy per unit volume is the triangular area under stress versus strain; the peak product alone is twice that area.

Why not the others: (A) halves the area twice; (C) adds an extra rectangle; (D) uses final stress over the whole range.

Key point: Read energy from the complete loading path.

Q50 / A Ice-water calorimetry with a phase change

Answer: (A) 17.5 °C.

Given: Hot water can release 300×4.2×50=63000 J by cooling to 0 °C; melting all ice needs 100×336=33600 J.

Formula/Principle: Test the phase-change threshold before balancing sensible heat.

Step 1: All ice melts, leaving 29400 J of heat.

Step 2: The 400 g resulting liquid warms by 29400/(400×4.2)=17.5 K.

Worked derivation:

$$Q_{\text{hot,max}} = 300(4.2)(50) = 63000\,\text{J} > Q_{\text{melt}} = 100(336) = 33600\,\text{J}; \quad T = \frac{63000 - 33600}{(300+100)4.2} = 17.5^{\circ}\text{C}.$$

Why: Temperature stays at 0 °C during melting; only excess energy raises the final temperature. The inequality first establishes that all ice melts, validating the final liquid-water heat-capacity step. If the hot water could not supply the latent heat, solving for a positive final temperature would be physically invalid.

Why not the others: (B) overlooks available heat; (C) neglects latent heat; (D) heats the wrong water mass.

Key point: Resolve phase change before using a single temperature-balance equation.

Q51 / D RMS-speed ratio at a common temperature

Answer: (D) 3:2.

Given: T is common; M_1:M_2=4:9.

Formula/Principle: vrms=√(3RT/M).

Step 1: $v_1/v_2=\sqrt{(M_2/M_1)}$.

Step 2: √(9/4)=3/2.

Worked derivation:

$$\frac{v_1}{v_2} = \sqrt{\frac{3RT/M_1}{3RT/M_2}} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{9}{4}} = \frac{3}{2}.$$

Why: At equal mean translational kinetic energy, lighter molecules move faster. Absolute temperature is shared, so a lighter molecule moves faster in the rms sense. Using molar mass directly without its inverse square root would reverse the result.

Why not the others: (A) uses masses directly; (B) fails to invert; (C) inverts but omits square root.

Key point: RMS speed is inversely proportional to square root of molar mass.

Q52 / C Period of series and parallel springs

Answer: (C) 2.

Given: kseries=k/2, kparallel=2k.

Formula/Principle: T=2π√(m/keffective).

Step 1: Ratio=√(kparallel/kseries).

Step 2: √[(2k)/(k/2)]=√4=2.

Worked derivation:

$$k_{\text{series}} = k/2, \quad k_{\text{parallel}} = 2k; \qquad \frac{T_s}{T_p} = \sqrt{\frac{m/(k/2)}{m/(2k)}} = 2.$$

Why: The series connection is softer and therefore oscillates more slowly. The same mass cancels, while the effective spring constants differ by a factor of four. Series springs each stretch under the full force; parallel springs split it.

Why not the others: (A) reverses the ratio; (B) changes only one spring configuration; (D) omits the square root.

Key point: Find equivalent stiffness before comparing periods.

Q53 / B Energy of disconnected series capacitors after dielectric insertion

Answer: (B) 2/3 and 1/2.

Given: Both series capacitors retain charge Q; C changes to 2C.

Formula/Principle: Utotal=ΣQ²/(2Ci), Vmodified=Q/Cmodified.

Step 1: Ui=Q²/(2C)+Q²/(4C)=3Q²/(4C).

Step 2: Uf=Q²/(4C)+Q²/(4C)=Q²/(2C), so Uf/Ui=2/3.

Step 3: Voltage changes Q/C→Q/(2C), a factor 1/2.

Worked derivation:

$$\frac{U_f}{U_i} = \frac{Q^2/(2C)}{3Q^2/(4C)} = \frac{2}{3}, \qquad \frac{V_{1,f}}{V_{1,i}} = \frac{Q/(2C)}{Q/C} = \frac{1}{2}.$$

Why: Source disconnection fixes charge, not voltage. The dielectric changes one capacitance, but no charge can enter or leave the disconnected series chain. Energy is not conserved during insertion because mechanical work and dielectric forces can exchange it.

Why not the others: (A) reverses energy ratio; (C) reverses voltage ratio; (D) assumes fixed energy.

Key point: Identify the conserved electrical quantity first.

Q54 / A Magnetic force with two field components

Answer: (A) qvB(k̂−ĵ).

Given: v=vî and magnetic field has equal +y and +z components.

Formula/Principle: F=q(v×B).

Step 1: î×ĵ=k̂, while î×k̂=−ĵ.

Step 2: Add component forces to obtain qvB(k̂−ĵ).

Worked derivation:

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} = qv\hat{\mathbf{i}} \times B(\hat{\mathbf{j}} + \hat{\mathbf{k}}) = qvB(\hat{\mathbf{k}} - \hat{\mathbf{j}}).$$

Why: The force is perpendicular to velocity and to the net field, not along either field component. Distribute the cross product before applying the right-hand cyclic order: i×j=k but i×k=−j. A dot-product calculation would give a scalar, not the force direction.

Why not the others: (B) reverses the cross-product order; (C) treats force as parallel to B; (D) ignores nonparallel motion.

Key point: Apply the right-hand rule separately to each component.

Q55 / D Potential change from a signed field graph

Answer: (D) Zero.

Given: The graph's positive rectangle has area $2E_0a$; the negative rectangle has area $-E_0(2a)=-2E_0a$.

Formula/Principle: $\Delta V=-\int E_x\,dx$, using signed area.

Step 1: The two signed field areas cancel.

Step 2: Therefore the negative of their sum is also zero.

Worked derivation:

$$V(3a) - V(0) = -\int_0^{3a} E_x \, dx = -[2E_0 a - E_0(2a)] = 0.$$

Why: Field reverses direction but remains nonzero; equal opposite potential contributions can cancel. The positive and negative rectangular signed areas exactly cancel. Potential responds to the signed integral, so the endpoint values can match even though the field is nonzero everywhere in the plotted regions.

Why not the others: (A) adds area magnitudes; (B) omits the second interval; (C) omits the first interval.

Key point: Retain signs and widths when reading an E–x graph.

Q56 / C Balanced Wheatstone bridge

Answer: (C) 6 Ω.

Given: Corresponding upper/lower resistance ratios must match for midpoint potentials to agree.

Formula/Principle: Bridge balance: 2/3=4/R.

Step 1: Cross-multiply 2R=12.

Step 2: R=6 Ω.

Worked derivation:

$$\frac{R_{\text{upper,left}}}{R_{\text{lower,left}}} = \frac{R_{\text{upper,right}}}{R_{\text{lower,right}}} \Longrightarrow \frac{2}{3} = \frac{4}{R} \Longrightarrow R = 6\,\Omega.$$

Why: Zero galvanometer current means no potential difference between its terminals, independent of the galvanometer's resistance. At null deflection no detector current flows, which makes the potential-divider ratios on the two branches equal. The bridge condition is a ratio, not equality of arm resistances.

Why not the others: (A) uses reciprocal ratio; (B) copies the left lower arm; (D) incorrectly adds arm resistances.

Key point: Match like-position arm ratios.

Q57 / B Lens and plane mirror on a common axis

Answer: (B) 50/3 cm left.

Given: Initial object distance 30 cm and lens-to-mirror distance 20 cm.

Formula/Principle: For each lens traversal 1/f=1/u+1/v; mirror images sit equally far behind a plane mirror.

Step 1: First image is v=15 cm right of lens, 5 cm in front of mirror.

Step 2: Mirror image is 5 cm behind it, 25 cm right of lens, now an object for returning light.

Step 3: Second lens traversal gives v=(10×25)/(25−10)=50/3 cm to the left.

Worked derivation:

$$\frac{1}{10} = \frac{1}{30} + \frac{1}{15}; \quad x_{\text{mirror}} = 2(20) - 15 = 25\,\text{cm}; \quad \frac{1}{10} = \frac{1}{25} + \frac{1}{v} \Longrightarrow v = \frac{50}{3}\,\text{cm}.$$

Why: The mirror changes propagation direction, so the second lens object is on the right. The mirrored intermediate point is 25 cm right of the lens; for light returning leftward it acts as the incident-side object. State the side as well as the positive magnitude to avoid a sign-convention ambiguity.

Why not the others: (A) stops at first lens; (C) stops at mirror image; (D) reverses final side.

Key point: Recalculate object and incident side after each optical element.

Q58 / A Threshold frequency from two stopping potentials

Answer: (A) f/2.

Given: $eV_1 = hf - \varphi = e(1\ V)$; $eV_2 = 2hf - \varphi = e(3\ V)$.

Formula/Principle: Subtract equations to eliminate work function: $hf = e(2\ V)$.

Step 1: $\varphi = hf - e(1\ V) = e(1\ V) = hf/2$.

Step 2: Threshold $f_0 = \varphi/h = f/2$.

Worked derivation:

$$e(3-1)\,\mathrm{V} = h(2f - f) \Longrightarrow hf = 2\,\mathrm{eV}; \quad \phi = hf - e(1\,\mathrm{V}) = 1\,\mathrm{eV}, \quad f_0 = \phi/h = f/2.$$

Why: The difference in stopping potentials measures the energy increase from doubling frequency. The difference of the two stopping-potential equations eliminates the unknown work function first. A stopping potential measures maximum kinetic energy per electron charge.

Why not the others: (B) uses the 3 V alone; (C) neglects nonzero kinetic energy at f; (D) places threshold above light already emitting electrons.

Key point: Eliminate φ with two measurements.

Q59 / D Bohr transition wavelength ratio

Answer: (D) 27/5.

Given: Energy gaps are $13.6(1/4 - 1/9) = 13.6 \times 5/36$ and $13.6(1 - 1/4) = 13.6 \times 3/4$.

Formula/Principle: Photon $\lambda = hc/\Delta E$; wavelength ratio is inverse gap ratio.

Step 1: $\lambda_{32}/\lambda_{21} = \Delta E_{21}/\Delta E_{32}$.

Step 2: $(3/4)/(5/36) = 27/5$.

Worked derivation:

$$\tilde{\nu}_{32} = R(1/2^2 - 1/3^2) = 5R/36, \quad \tilde{\nu}_{21} = R(1 - 1/2^2) = 3R/4; \quad \frac{\lambda_{32}}{\lambda_{21}} = \frac{\tilde{\nu}_{21}}{\tilde{\nu}_{32}} = \frac{27}{5}.$$

Why: The 3→2 transition has a smaller energy gap and hence longer wavelength. The smaller 3→2 energy difference means a longer photon wavelength. Wavelength ratios invert the corresponding wavenumber ratios; level-number ratios themselves do not determine spectra.

Why not the others: (A) reverses ratio; (B) and (C) use only part of the level fractions.

Key point: Compute energy gaps before taking inverse wavelength ratios.

Q60 / C Lenz law and induced-current direction

Answer: (C) Counterclockwise, producing outward field.

Worked derivation:

$$\Delta\Phi_{\text{into}} > 0 \implies \mathbf{B}_{\text{induced}} \text{ points outward}; \qquad \text{counterclockwise current gives outward field.}$$

Why: The induced effect opposes the *increase* of inward flux, requiring an outward magnetic field. Right-hand fingers curling counterclockwise give a thumb pointing out of the page. Fixed field direction does not mean fixed flux when magnitude changes. Apply Lenz's law to the *change* in flux, then apply the right-hand rule to the induced field. This sequence avoids confusing field direction with whether the field magnitude is rising.

Why not the others: (A) Reinforces the increase. (B) Clockwise current produces inward field. (D) Ignores changing magnitude.

Key point: Determine the opposed flux change before using the right-hand rule.
